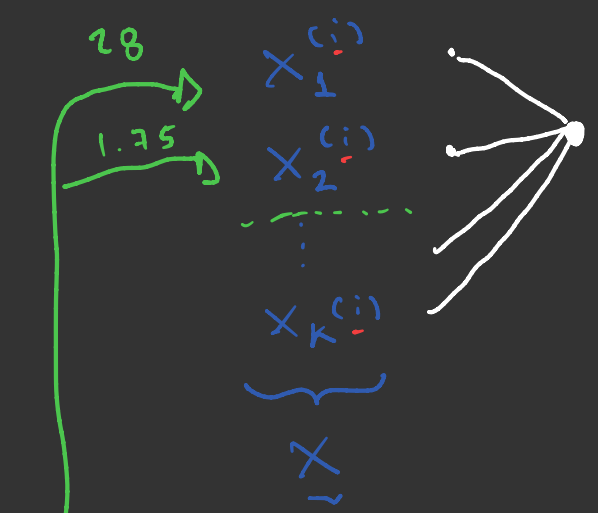


Perceptron Simple

$$\varphi(\underline{x}, \underline{\beta}^*)$$

logístico

$$\sigma(z) = \frac{e^z}{1 + e^z}$$



$$\hat{y} = \varphi(\underline{x}, \underline{\beta}^{(t)}) = z$$

$$(1, \underline{x}_1^{(i)}, \underline{x}_2^{(i)}, \dots, \underline{x}_k^{(i)})$$

$$\begin{pmatrix} \beta_0^{(t)} \\ \beta_1^{(t)} \\ \beta_2^{(t)} \\ \vdots \\ \beta_k^{(t)} \end{pmatrix}$$

	$\underline{x}$		y	$\hat{y}$
1	23	1.73	2.5	
2	25	1.72	2.8	
i	28	1.75	2.2	
n	30	1.8	2.1	
	x_1	x_2		

$$\underline{\dot{x}} \cdot \underline{\beta} = z$$

(Predictor)
linear

$$= \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \dots + \beta_k x_k$$

Modelo lineal

$$\varphi(z) = z$$

Ejemplo 1. Percepción lineal simple

X		y	z
x ₁	x ₂		
18	45	5.1	5.245
19	52	4.8	3.824
20	48	4.9	4.416
21	55	4.2	
22	52	4.1	
23	60	3.5	
25	57	3.8	
26	63	3.2	
30	60	3.1	
31	70	2.5	
40	62	2.5	
41	68	1.8	
50	68	2.1	
51	64	1.9	

$$\hat{y} = z$$

$$e = y - \hat{y}$$

$$\beta_0 = 16$$

$$\beta_1 = -0.14$$

$$\beta_2 = -0.183$$

$$\begin{matrix} -0.145 \\ 0.976 \\ 0.484 \end{matrix}$$

$$1.9$$

$$5.1$$

$$\beta_1 x_1$$

$$\beta_2 x_2$$

$$\beta_0 \quad x_1 \quad \beta_1$$

$$x_2 \quad \beta_2$$

$$z = 16 + 19 \cdot (-0.14) + 52 \cdot (-0.183)$$

$$3.824$$

$$z = 16 + (-0.14)x_1 + (-0.183)x_2$$

$$\beta^{(0)} = (\beta_0^{(0)}, \beta_1^{(0)}, \beta_2^{(0)})$$

$$\beta_0^{(0)} = 47.2 / 14 = 3.371 \quad \times 2 = 6.742$$

$$\beta_1^{(0)} = (5.1 - 1.8) / (18 - 41) = 3.3 / -23 = -0.14 \quad \checkmark$$

$$\beta_2^{(0)} = (5.1 - 1.8) / (45 - 68) = 3.3 / -18 = -0.183 \quad \checkmark$$

$$(y_a - y_b) / (x_{1a} - x_{1b})$$

$$(y_a - y_b) / (x_{2a} - x_{2b})$$

$$\beta^{(0)} = \begin{pmatrix} 3.371 \\ -0.14 \\ -0.183 \end{pmatrix}$$

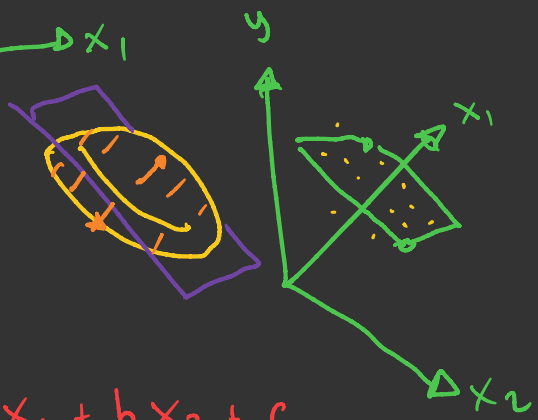
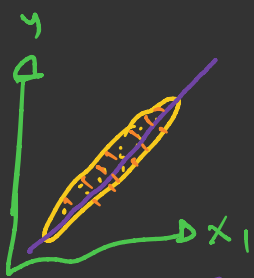
$\beta_0^{(0)}$ $\beta_1^{(0)}$ $\beta_2^{(0)}$
 x_1 x_2

$$z = (1 \quad x_1 \quad x_2) \begin{pmatrix} 6.742 \\ -0.14 \\ -0.183 \end{pmatrix}$$

$$\begin{aligned} (1 \quad 18 \quad 45) \begin{pmatrix} 6.742 \\ -0.14 \\ -0.183 \end{pmatrix} &= 1 \cdot 6.742 + 18(-0.14) + 45(-0.183) \\ &= 6.742 - 2.52 - 8.235 = -4.293 \\ &\downarrow 6 \\ &= 5.245 \end{aligned}$$

$$\hat{y} = \beta_0 + \beta_1 x$$

$$y = mx + b$$



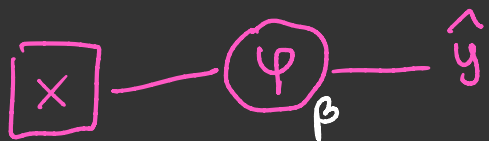
$$y = ax_1 + bx_2 + c$$

$$\hat{y} = \beta_0 + \beta_1 x_1 + \beta_2 x_2$$

z

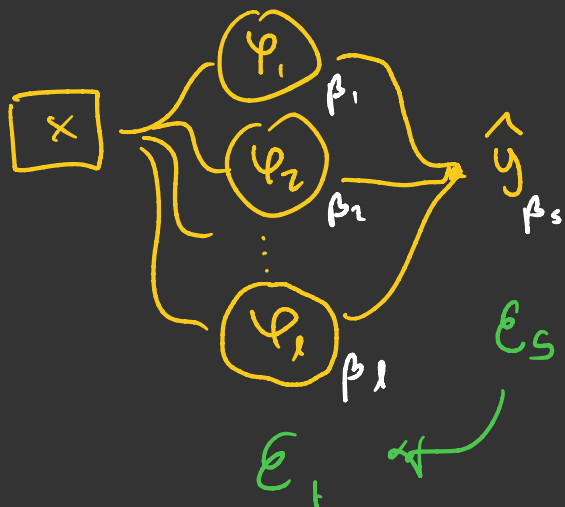
$$L = \frac{1}{2} \epsilon^2$$

Percepción



$$\underline{\beta^* = \beta + \eta \nabla_{\beta} \hat{y}}$$

Red neuronal



$$\beta_s^* = \beta_s + \eta \mathcal{J}_{\beta_s} \epsilon_s$$

$$\beta_1^* = \beta_1 + \eta \mathcal{J}_{\beta_1} \epsilon_1$$

$$\beta_2^* = \beta_2 + \eta \mathcal{J}_{\beta_2} \epsilon_2$$

⋮

$$\beta_l^* = \beta_l + \eta \mathcal{J}_{\beta_l} \epsilon_l$$

$$\epsilon_s = y - \hat{y}$$

$$\epsilon_1 = \mathcal{J}_{h_1^{(1)}} \epsilon_s$$

$$\epsilon_2 = \mathcal{J}_{h_2^{(1)}} \epsilon_s$$

⋮

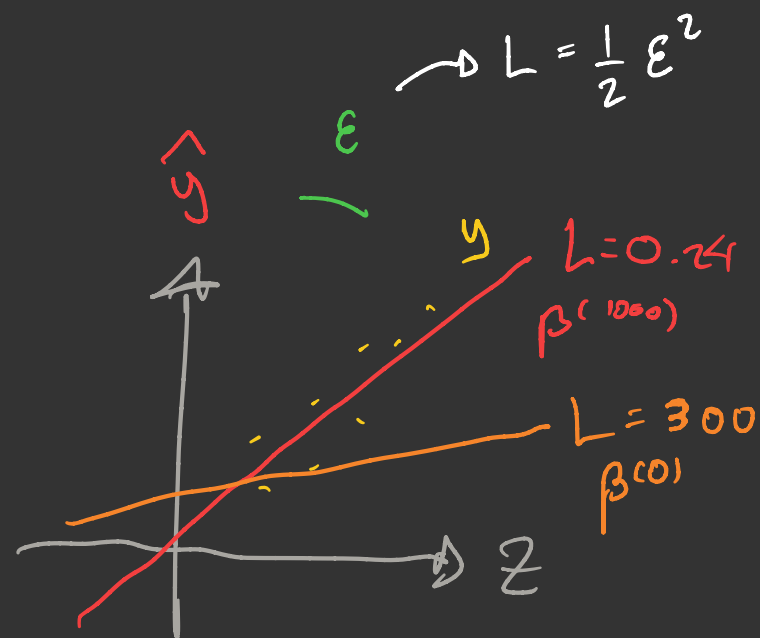
$$\epsilon_l = \mathcal{J}_{h_l^{(1)}} \epsilon_s$$

2 - características

$\beta_0^{(0)}$	$\beta_1^{(0)}$	$\beta_2^{(0)}$	y
18	42	5.1	
19	52	4.9	
20	45	4.8	
21	50	4.2	

1 - característica

$\beta_0^{(0)}$	$\beta_1^{(0)}$	y
18	5.1	
19	4.9	
20	4.8	
21	4.2	



Percepción $\hat{y} = \varphi(\underline{x}, \underline{\beta})$

lineal $\underline{x} \cdot \underline{\beta}$
logístico $\sigma(\underline{x} \cdot \underline{\beta})$

$$\beta^{(1)} = \beta^{(0)} + \eta \nabla_{\beta} L$$

β^*

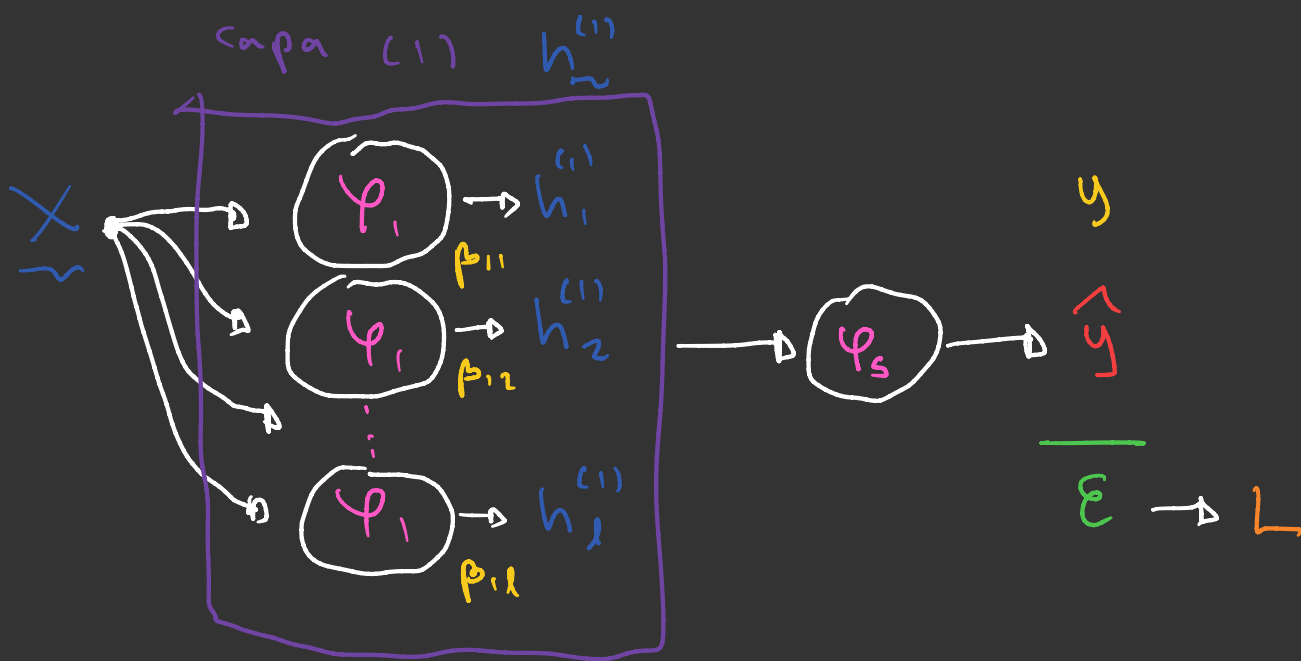
Red neuronal de una capa

$$\hat{y} = \varphi_s(\underline{h}^{(1)}, \beta_s)$$

$$\underline{h}^{(1)} \begin{cases} h_1^{(1)} = \varphi_1(\underline{x}, \beta_{11}) \\ h_2^{(1)} = \varphi_1(\underline{x}, \beta_{12}) \\ \vdots \\ h_l^{(1)} = \varphi_1(\underline{x}, \beta_{1l}) \end{cases}$$

$$\nabla_{\beta_1} \rightarrow J_B$$

$\geq <$



$$\underline{x} \rightarrow \varphi_1(\underline{x}, \underline{\beta}) \rightarrow \varphi_s(\underline{h}^{(1)}, \beta_s) \rightarrow \hat{y}$$

Percepción $\rightarrow \beta_0, \beta_1, \beta_2, \dots$

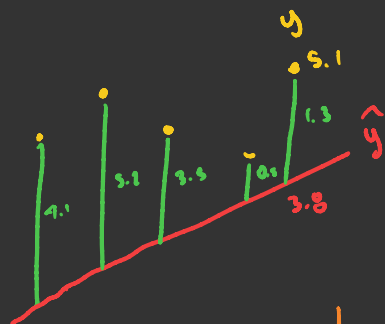
$$\varphi(\underline{x}, \beta_n^{(0)}) = \hat{y} - y$$

$$\mathcal{E} = y - \hat{y}$$

Modelo de
activación / respuesta

1-D ∇
2-D Hessiano

$\propto \begin{cases} \text{Newton} \\ \text{BFGS} \end{cases}$



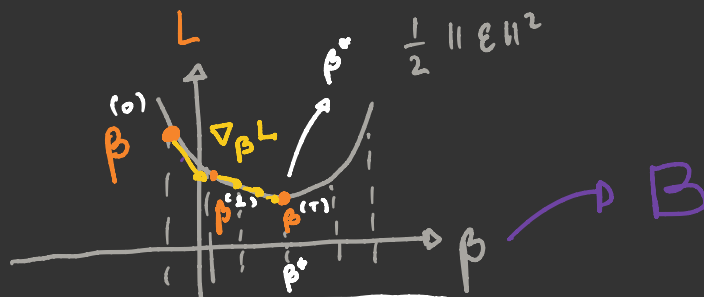
$$\mathcal{E} = (4.1, 3.2, 3.5, 0.5, 1.3)$$

$$L = \frac{1}{2} \mathcal{E}^T \mathcal{E}$$

$$\frac{1}{2} \|\mathcal{E}\|^2$$

$$\nabla(\partial_{\beta_0}, \partial_{\beta_1}, \dots)$$

$$J \begin{pmatrix} \partial_{\beta_{11}} & \partial_{\beta_{12}} \\ \partial_{\beta_{21}} & \partial_{\beta_{22}} \\ \partial_{\beta_{31}} & \partial_{\beta_{32}} \end{pmatrix}$$



$$\beta^{(1)} = \beta^{(0)} + \eta \nabla_{\beta} L$$

Optimización
por el descenso
del gradiente

$$\Delta J_{\beta} L$$